# UNIVERSIDAD NACIONAL DE INGENIERÍA

## Facultad de Ciencias

### Escuela Profesional de Ciencia de la Computación

# EXAMEN FINAL

Curso: **Fundamentos de Programación CC112 - A**

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1.

Para n = 3 (filas) , m = 4 (columnas)

```
test01@pcs5:~/pc02-cc112a-albganviniu-design/semana8/examenfinal$ valgrind ./build/debug/pregunta1
==13059== Memcheck, a memory error detector
==13059== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==13059== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==13059== Command: ./build/debug/pregunta1
==13059==
Ingrese el numero de filas: 3
Ingrese el numero de columnas: 4

Matriz:
1       2       3       4
2       4       6       8
3       6       9       12

Suma de cada fila:
Fila 1: 10
Fila 2: 20
Fila 3: 30
==13059==
==13059== HEAP SUMMARY:
==13059==     in use at exit: 0 bytes in 0 blocks
==13059==   total heap usage: 8 allocs, 8 frees, 75,860 bytes allocated
==13059==
==13059== All heap blocks were freed -- no leaks are possible
==13059==
==13059== For lists of detected and suppressed errors, rerun with: -s
==13059== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
test01@pcs5:~/pc02-cc112a-albganviniu-design/semana8/examenfinal$
```

Para n = 4 (filas), m = 3 (columnas)

```
test01@pcs5:~/pc02-cc112a-albganviniu-design/semana8/examenfinal$ valgrind ./build/debug/pregunta1
==13100== Memcheck, a memory error detector
==13100== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==13100== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==13100== Command: ./build/debug/pregunta1
==13100==
Ingrese el numero de filas: 4
Ingrese el numero de columnas: 3

Matriz:
1       2       3
2       4       6
3       6       9
4       8       12

Suma de cada fila:
Fila 1: 6
Fila 2: 12
Fila 3: 18
Fila 4: 24
==13100==
==13100== HEAP SUMMARY:
==13100==     in use at exit: 0 bytes in 0 blocks
==13100==   total heap usage: 9 allocs, 9 frees, 75,872 bytes allocated
==13100==
==13100== All heap blocks were freed -- no leaks are possible
==13100==
==13100== For lists of detected and suppressed errors, rerun with: -s
==13100== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
test01@pcs5:~/pc02-cc112a-albganviniu-design/semana8/examenfinal$
```

2.

```
test01@pcs5:~/pc02-cc112a-albganviniu-design/semana8/examenfinal$ valgrind ./build/debug/pregunta2
==13407== Memcheck, a memory error detector
==13407== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==13407== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==13407== Command: ./build/debug/pregunta2
==13407==

PUNTO 1:
Ingrese la coordena 'X' del punto: 2.4
Ingrese la coordenada 'Y' del punto: 8.5

PUNTO 2:
Ingrese la coordena 'X' del punto: 12.5
Ingrese la coordenada 'Y' del punto: 21.7

El punto mas lejos del origen es el PUNTO 2 con las coordenadas:
X: 12.5         Y: 21.7

==13407==
==13407== HEAP SUMMARY:
==13407==     in use at exit: 0 bytes in 0 blocks
==13407==   total heap usage: 9 allocs, 9 frees, 76,036 bytes allocated
==13407==
==13407== All heap blocks were freed -- no leaks are possible
==13407==
==13407== For lists of detected and suppressed errors, rerun with: -s
==13407== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
```

P1(3, 4) ; P2(-3, -5)

```
test01@pcs5:~/pc02-cc112a-albganviniu-design/semana8/examenfinal$ valgrind ./build/debug/pregunta2
==13514== Memcheck, a memory error detector
==13514== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==13514== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==13514== Command: ./build/debug/pregunta2
==13514==

PUNTO 1:
Ingrese la coordena 'X' del punto: 3
Ingrese la coordenada 'Y' del punto: 4

PUNTO 2:
Ingrese la coordena 'X' del punto: -3
Ingrese la coordenada 'Y' del punto: -5

El punto mas lejos del origen es el PUNTO 2 con las coordenadas:
X: -3           Y: -5

==13514==
==13514== HEAP SUMMARY:
==13514==     in use at exit: 0 bytes in 0 blocks
==13514==   total heap usage: 9 allocs, 9 frees, 76,036 bytes allocated
==13514==
==13514== All heap blocks were freed -- no leaks are possible
==13514==
==13514== For lists of detected and suppressed errors, rerun with: -s
==13514== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
test01@pcs5:~/pc02-cc112a-albganviniu-design/semana8/examenfinal$
```